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PAPER

Biofabrication of spheroids fusion-based tumor models: computational simulation of glucose effects

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E-mail: nimoldov@iupui.edu**Keywords:** spheroids, tumor, agent-based modeling, tissue engineering, biofabrication, diabetes, scaffold-free bioprintingSupplementary material for this article is available [online](#)

Abstract

In vitro tumor models consisting of cell spheroids are increasingly used for mechanistic studies and pharmacological testing. However, unless vascularized, the availability of nutrients such as glucose to deeper layers of multicellular aggregates is limited. In addition, recent developments in cells-only biofabrication (e.g. ‘scaffold-free bioprinting’), allow the creation of more complex spheroid-based structures, further exposing the cells to nutrient deprivation within these constructs. To explore the impact of glucose availability on such tumor-like structures, we used the CompuCell3D platform for modeling of tumor spheroids. By monitoring the types of cells, fusing pairs geometry and the distance between spheroids centers of mass, we made novel heuristic observations on how binary- and multi-spheroid fusions are impacted by glucose availability. At limiting glucose concentrations mimicking hypoglycemia we noted an abrupt collapse of the tumor spheroids, unexpectedly amplified by the contact with normal cell spheroids. At higher glucose concentrations, we found an increased intermixing of cancerous cells, strong anti-phase oscillations between proliferating and quiescent tumor cells and a structural instability of fusing tumor spheroids, leading to their re-fragmentation. In a model of tumor microenvironment composed of normal cell spheroids fusing around a tumoral one, the competition for glucose lead to either the tumor’s disappearance, to a steady state, or to its expansion. Moreover, the invasion of this microenvironment by individual tumor cells was also strongly depended on the available glucose. In conclusion, we demonstrate the value of computational simulations for anticipating the properties of biofabricated tumor models, and in generating testable hypotheses regarding the relationship between cancer, nutrition and diabetes.

1. Introduction

The tumors manifest subtle structural properties which impact their development and response to treatment. In particular, intra-tumor distribution of nutrients and waste products is related in a complex manner to their size, cellular composition,

metabolism, and vascularization, and profoundly impacts on tumor’s properties and behavior. This relationship is further complicated by metabolic disorders of the patient, such as diabetes, notorious for its wide fluctuations in glucose and insulin concentrations. Moreover, if the tumors develop in patients who are taking anti-diabetic medication, this may

add another layer of complexity. Therefore, the relationship between diabetes and tumor development has remained insufficiently understood so far. Computational models could help simplify this analysis, provide heuristic insights, and underscore relationships otherwise difficult to establish between the key variables [1].

Among the methods to create *in vitro* tumor models are those based on biofabrication [2], including bioprinting [3]. Some of these methods revolve around the use of cell spheroids as clusters of cells formed by self-assembling, with the contribution of cellular turnover, i.e. cell proliferation and cell death [4]. In the absence of supporting biomaterials for construct biofabrication, these are named ‘scaffold-free’ methods [5].

In general, the spheroids are formed and maintained through the general (bio)physical principle of the system’s energy minimization, by maximal engagement of intercellular adhesion molecules (mainly cadherins) [6]. The same principle applies when spheroidal aggregates come in contact, leading to their fusion into larger 3D structures [7]. The elementary step is the *fusion* of two spheroids of normal and/or cancer cells, which for its fundamental role has been directly studied [8, 9]. This mechanism to generate cellular structures of various desirable geometries was recognized in the early phase of tissue engineering field, when the use of cell spheroids as building blocks for the ‘scaffold-free’ (i.e. hydrogel bioink-independent) form of bioprinting was suggested [7]. This fusion is also the key step in the ‘Kenzan’ (microneedle-assisted) bio-fabrication/bioprinting method [10]. Naturally, this method can be useful to create tumor models as well [4].

Tumor cell spheroids (or ‘tumor nodules’, also simply called here ‘tumors’, to distinguish them from the spheroids prepared from normal cells) were previously modeled *in silico*, albeit not their fusions yet, by various methods [1, 11]. Among these is CompuCell3D (CC3D), an open-source agent-based modeling (ABM) platform of the ‘cellular Potts’ modeling class [12]. This is particularly suited to capture cellular dynamics, as well as the biochemical [13], metabolic [14], genetic and/or biophysical [15] factors controlling cell behavior, while ignoring other cellular properties (e.g. subcellular structure or biomaterial properties) [16]. Using this platform, many other biological processes were insightfully modeled, such as tumorigenesis [17], embryonic development [18], cyst formation [15], neovascularization [19], and others. Moreover, a detailed description of tumor spheroid formation and stability has been developed in CC3D, with a focus on the role of available glucose on a single tumor’s cellular self-organization and stability [14].

Here we used CC3D modeling to address the impact of available glucose on spheroid assembling dynamics in larger 3D structures, starting with their

binary fusion. This approach expands upon the computational model developed by Swat *et al* [14], for capturing the behavior of tumor spheroids in CC3D, which uses the Glazier–Graner–Hogeweg (GGH) ABM principles [20]. This method addresses cell behavior and development through a Monte Carlo simulation approach [11, 21]. The original model described the behavior of single tumor spheroids through several independent parameters, including nutrient (glucose) availability [14]. Ours is a modification of this model, adapted to describe both tumor and ‘normal’ cell spheroids, and their fusion, contingent on available glucose. Although the normal spheroids [18] and their fusion [21] have been previously modeled by the GGH approach, the role of glucose in this process, particularly when involving fusing tumoral spheroids, has not been considered yet.

Our heuristic computational modeling was mainly intended to explore the landscape of possible outcomes of spheroids fusion, as dependent on the available glucose. Using it, we identified remarkable cell-behavioral patterns worth of further computational and experimental exploration. Besides biofabrication and tissue engineering implications, the results presented here could be also relevant for the impact of diabetes on cancer progression, a problem of large medical importance [22, 23].

2. Methods

2.1. The computational model and the ‘cell types’

We used the open-source CC3D code for modeling tumor spheroids published in Swat *et al* [14], with some modifications. To describe the cellular composition of tumors, in that study were defined five different types of cells considered: (a) proliferative cells, (b) quiescent cells, (c) necrotic cells, (d) proliferative stem cells, and (e) quiescent stem cells. Moreover, to maintain the discrete treatment of the space, and to allow its re-distribution for modeling purposes, in the model there is also a 6th type of unit (representing a discrete ‘medium’), as the environment the other cell types exist in, that mimics some of the properties of the tumor stroma and of its extracellular matrix [14].

Assuming their role in the fusion of tumor spheroids (which was our main focus) as being limited, and to simplify the computation, the stem cells of both types (d) or (e) were not actively used in our model. However, instead of eliminating them as variables, they were only ‘disabled’; that is, through the initialization of our model (as described later), we averted the possibility of stem cells being created by mitosis, or by the evolution of tumor cells [14]. Moreover, to implement the cells needed for modeling the ‘normal’ cell spheroids, we reverse-engineered normal cells from the tumor cell phenotype by ‘muting’ (setting to zero) in the CC3D model the parameters related both to their stemness and to their proliferation. Indirectly, this also zeroed a stochastic cadherin mutation rate

operating in the original model at every cell division step [14]. Thus, our approach helps avoiding other subtle and hard to control differences that may have arisen when developing the normal and cancer cell models independently.

2.2. Cell adhesion-related parameters

The original model assigned variables called ‘cadherins’, ‘integrins’, and ‘fibronectin’ as parameters associated with each cell type that determine the strength of binding interaction between cells and with their environment, represented by the ‘medium’ as defined above. Naturally, only the medium presents ‘fibronectin’, which our model also follows, and with the same relative expression density as in [14] (set to a value of 16). The relative cadherin expression in cancer cells which was originally set to 8, was also held unmodified in our version of the model. To increase the spheroids’ stability, in the normalized (‘muted’) cells, we hypothesized the relative strength of cadherin-based interaction to be double than that of the cancer cells, and thus we gave this parameter a value of 16. Moreover, to determine the impact of this parameter on the continence of our boxed spheroid model of tumor microenvironment, we also performed a limited ‘parameter scan’ for normal cell cadherin, giving it values of 9, 12, and 15.

Regarding the energy of interaction, the relative adhesive terms were defined in the original model to describe the binding strength for cadherin–cadherin and integrin–fibronectin interactions. In our model, the strength of cadherin-to-cadherin binding constant was held the same, with a value of 2. Also, we did not eliminate the probability for mutations to occur to cadherin density: that variable was maintained at 10% chance per mutation, but cadherin-to-cadherin binding strength relative to that of integrin-to-fibronectin was changed from 0.2 to 0.02.

2.3. Other cell parameters

Swat *et al* [14] provided a list of additional parameters for the model, which remained unmodified here, except; (a) growth rate, defined as pixels/Monte Carlo step (MCS) of ‘muted’ cells in our generalized ‘normal’ cell, was set to zero (as expected for a stable, quiescent tissue); (b) for the same reason, the ‘maximum number of divisions’ allowed for normal cells was also set to zero; (c) consequently, as the division was impossible, the probability of a normal cell to differentiate into a stem cell after division was also set to zero.

Another feature of the original model was that at every MCS in the simulation, each individual cell’s glucose concentration is evaluated at its center: if this concentration is above a threshold, the cell grows and remains healthy, but if it is below this threshold it will accumulate a ‘point’ of damage. When it accumulates enough such damage points, the cell will irreversibly turn necrotic and die off. However, for those

cells which manage to grow to twice their size and to divide into two daughter cells, the ‘damage counter’ is reset to zero points in both daughter cells. In our case, to compensate for the inability of the ‘muted’ cells to reset their damage counter through division, the total allowed ‘damage’ points before onset of necrosis was increased ten times (from 102 to 1020), a finite value chosen to leave the opportunity for cell death in case of starvation.

2.4. Steady-state glucose diffusion field

In the CC3D tumor model [14], the glucose diffusion field was described through: (a) diffusion constant; (b) decay constant; (c) secretion constant, and (d) cell uptake rate. These parameters were taken from experimental values, such that an equilibrium of glucose concentration could be reached in the medium when no cells were present. In our model, the decay constant, secretion constant, and cell uptake rates remained consistent with the original model. However, we made the diffusion constant an independent variable, and varied it from ten times lower to ten times larger than ‘physiologic’ (herein also called ‘normal’) concentrations based on [14].

2.5. Spheroid initialization and placement

To analyze spheroids growth and changing composition, the Swat *et al* model used a single tumor stem cell to initialize them [14]. Here we maintained a similar approach to implement entire spheroids, but ‘muted’ the normal ones as described, when they reached 15 pixels in radius.

Of note, although the program is called CC3D, for computational reasons in practice it considers as ‘cells’ flat domains composed of pixels (rather than voxels), which by energy minimization generate 2D disks (rather than true 3D ‘spheres’) [14].

Here, we initiated *pairs* of spheroids of varying combinations: (a) muted with muted (also called ‘normal’ doublets), (b) muted with cancerous (‘heterogenous’ doublets), and (c) cancerous with cancerous (‘homogenous’ doublets) spheroids. For all fusing spheroids, we placed their centers 32 pixels apart, and thus there were initially two pixels left between them. However, with the width of each individual cell being three pixels, the very first iteration step caused them to touch. Of note, because in muted cells the mitosis feature was blocked (but—for convenience—not the initial cell growth), there is a non-negligible increase in the size of each cell in the normal spheroids between 0 and 100 MCS, visible both in images, and in graphs.

2.6. Tracking binary spheroid fusion by the distance between their centers of mass (DCM)

Because tumor spheroids doublets could become quite irregular during fusion, here we introduced a different method to monitor it, which does not rely on object’s shape. Since the center of mass of every cell in each spheroid can be tracked by the CC3D program,

the individual coordinates of every cell were recorded and averaged to produce a single overall x' and y' coordinate for each spheroid. Then these values were inserted into a formula of 'Distance between Centers of Mass' (DCM):

$$\text{DCM} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (1)$$

where the indices '1' and '2' refer to the overall center of mass coordinate for the first and second spheroid, respectively.

2.7. Monitoring binary spheroid fusion by measures of pair geometry

Another method to track spheroids fusion is to measure the lengths and widths of the fusing pairs [9]. To this end, we took screenshots of the simulations at 100 MCS intervals up to 15 000 MCS (in triplicate), which ensured that each spheroid pair reached at least 99% fusion. These image sets were compiled into AVI files at 2400×2400 ppx resolution. We then used a custom-written applet in Python, which first unrolls each frame in their respective AVI. The length and width of the doublet in any given frame was determined by scanning the images vertically then horizontally, to find the maximum, contiguous, non-medium pixel set for each row (corresponding to maximum Feret diameter) and column (minimum Feret diameter), respectively [9]. These length and width measurements were then stored with respect to their MCS value. The pixel measurement for each doublet was then normalized by taking the ratio of a given value of length or width in the course of simulation, over the corresponding initial value (at 0 MCS). The mean values for a given MCS from the triplet were then calculated, as well as the standard deviation. The statistical significance of differences between the normal and tumor case was assessed by *t*-test.

In addition, we introduced an estimation of the doublets 'aspect ratio' by dividing their length to the width, and also represented this variable as dependent on time of simulation and glucose availability.

2.8. Tumor model with cellular microenvironment

To simulate the more complex contribution of the cellular microenvironment to tumor biology, we generated 3×3 arrays of spheroids having a tumor nodule in the center and surrounded by eight normal cell spheroids, all in initial marginal contact, as in a real tissue-engineered system [4].

The diffusive migration of the tumor cells in this microenvironment at each step was quantified by overlying over the construct a dynamically sized circle, determined by the average radius of the tumor spheroid, with an area equal to respective step's total tumor cell area and the center placed at tumor's center of mass. The tumor cells outside this circle area were counted and displayed as the percentile fraction of the total, as dependent on time, glucose influx rate

outside the construct, and cadherin values. For a cell adhesion parameter scan in these constructs, the cadherin value for the tumor cells was maintained to 8 as in [14], while those of normal cells varied in the range 9–15 (to account for the fact that metastatic tumor cells are in general less adhesive than the normal ones [24]).

3. Results

3.1. Fusion of cell spheroids at normal glucose concentrations

It has been previously shown both experimentally and computationally with CC3D [25] that, due to the physics of the process which is mainly driven by the system's energy minimization [26], the placement in contact of two normal cell spheroids leads to their eventual amassing into a larger, round and stable spheroid. Here we found the same outcome with the normal cell spheroids using the modified CC3D program (i.e. after 'muting' their cancerous features) (figure 1(A), top). Also, the higher local cell density during and after fusion has stably and locally reduced the glucose concentrations (figure 1(B), top).

However, when we interacted normal with tumoral spheroids, their fusion proceeded with several new, unanticipated features (figure 1(A), middle). First, although they had the same density of surface adhesion molecules, the two cell populations did not readily mix, the normal cells remaining more aggregated than their transformed counterparts, possibly due to the latter's higher mobility, increased by the active cell division. Tracking the cell types in the fusing spheroids (figure 1(C)) helped us better understand this cell dynamics. Although the normal glucose level was selected to provide the non-tumoral cells enough metabolic supply to maintain the spheroids during the whole fusion process 'healthy' (i.e. with no cell death), when brought in contact with a tumor spheroid, the normal cells started to progressively die and were removed from the construct; for this reason, the number of 'normal necrotic cells' did not increase (figure 1(C), middle). Regarding the tumor cells in the fusing hybrid spheroids, after a phase of oscillatory instability their number increased quasi-linearly, not due to the dividing/quiescent proportion (which remains constant, figure 1(C), middle), but to the accumulation of the necrotic tumor cells. Surprisingly, these necrotic cells were mixed with the other tumor cells at random in a poorly shaped, lobular construct, and not located at its core, which remained occupied by normal cells (figure 1(A), middle). However, such a central distribution of necrotic tumor cells did occur in the structure generated by the fusion of tumor spheroids. Moreover, the structural instability noted before here became so severe that it led to breaking of the structure back in two separated parts (figure 1(A), bottom).

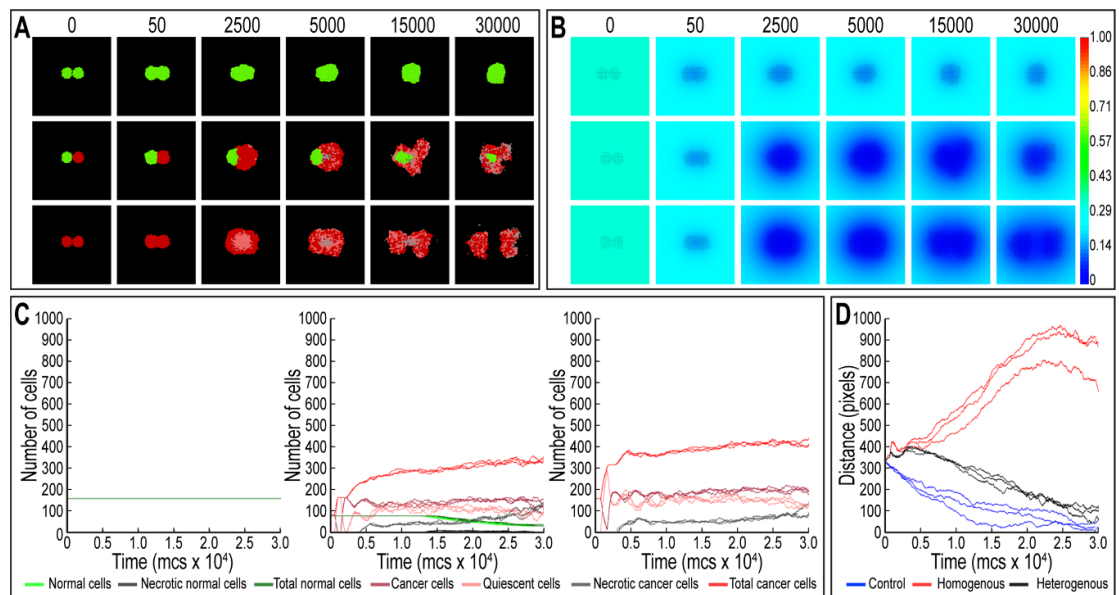


Figure 1. Fusion of cell spheroids at normal glucose concentrations. (A) Stages of fusion between two normal (upper row), heterogenous pair (central row) and two tumoral spheroids (lower row), at the indicated MCS steps. (B) Changes of glucose levels in the media during spheroids fusion (placed as in A); color of background indicates the relative change of glucose concentrations relative to the initial values (set to 0.50 on the color scale). (C) Kinetics of cell subpopulations in the fusing complex (left: normal spheroids; middle: heterogenous pair; right: homogenous tumor spheroids) (color code as in A). (D) Monitoring spheroids fusion via the distance between their centers of mass (new color code specified in the insert; results of triplicate runs for each condition).

To monitor spheroids fusion, here we introduced as a new method the tracking of distance between the initial spheroids' center of mass (DCM), and found as expected its continuous reduction in the normal case (figure 1(D), blue); the curve shapes for the three replicates were similar, although their trajectories differed slightly, likely due to the random component in cell mixing during fusion. As the tumoral spheroids tended to 'engulf' the normal ones (which could also shrink, due to nutrient deprivation at the core of hybrid spheroid, as shown in figure 1(A), middle), this caused the DCM to first increase (due to tumor cell proliferation) and then to decrease (due to fusion and/or engulfment), albeit at a slower rate than during fusion of normal spheroids (figure 1(A)). Thereafter, the tumorous portion of the adduct began to also shrink, or to undergo a slower expansion, a pattern consistent at all glucose concentrations and for all tested heterogenous mixing situations (below).

The size of fusing homogenous pairs of tumor spheroids in the same conditions and time points continued to expand (figure 1(A), bottom). The fusion kinetics of these tumor nodules was again different from the previous cases, as monitored by snapshots taken during simulations, but also through their DCM (figure 1(D)): the initial expansion, displaying a slight oscillatory pattern of DCM (found in all instances involving tumor cells, and correlating with a similar one in the cell counting graphs), was followed by a substantial increase accompanied by larger fluctuations in the triplicates, likely due to the stochastic nature of the process. This kinetics combines the

positive balance of cell proliferation vs. cell death (manifested mainly at the tumoral cores) with the intrinsic structural instability, leading to the reversal of these spheroids' association, with frequent binary fragmentation of the system (figure 1(A), bottom, and in the subsequent hyperglycemic conditions). However, the origin of this structural instability must have been purely dynamical, since it did not seem to derive from the distribution of glucose field itself (which remained relatively symmetrical throughout the simulation in figure 1(B), middle and bottom), and a triggering event could not be found in the DCM graphs, or by tracking the cellular subpopulations which showed smooth paths (figure 1(C), right and figure 1(D)).

3.2. Fusion of spheroids in hypoglycemic conditions

Reducing the available glucose in our model by manipulating the rate of its generation, had concentration-dependent effects on spheroids fusion. While the 50% reduced glucose rate generation only decreased cell numbers secondary to inhibition of cell divisions (figure S1 (available online at stacks.iop.org/BF/13/035010/mmedia)), between this and $0.2 \times$ (20%) glucose generation there was a more substantial decrease in the size of the fusing tumor nodules, as well as in that of normal spheroids (figure S2(A)). This reduction was the result of an abrupt decrease in nutrient supply early on, as soon as 50 MCS, produced by the fusion itself (compare figures S2(A) and (B)). This 'metabolic shock' reverberated later by affecting the 'damage' threshold of

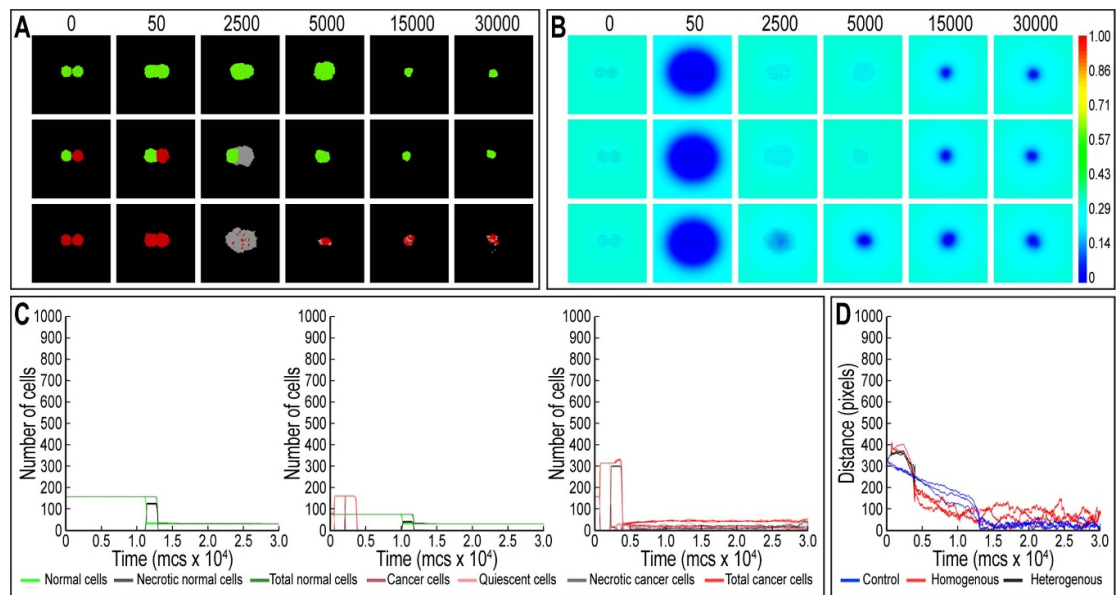


Figure 2. Fusion of cell spheroids at $0.1\times$ glucose concentration. (A) Snapshots of fusion between two normal (upper row), heterogenous (central row) and two tumoral spheroids (lower row). (B) Changes in the corresponding glucose fields. (C) Kinetics of cell subpopulations in the fusing complex (left: normal spheroids; middle: heterogenous pair; right: tumor spheroids) (color code as in A). (D) Monitoring spheroids fusion via the distance between their centers of mass.

the cells and leading to their death (figure S2(C)). A cooperative effect was detectable at about 400 MCS, when there was a precipitous decrease of DCM, corresponding to a collapse of these tumoral spheroids (figure S2(D)).

In the ‘starving’ $0.1\times$ condition (10% glucose compared to control), even normal spheroids underwent size reduction (figure 2(A), upper) also due to the abrupt decrease in glucose triggered by the fusion (figure 2(B) at 50 MSC) and leading to their stabilization at constant lower numbers (figure 2(C), left). This stability was catastrophic to the tumor spheroid in the hybrid combination which entirely died off (figure 2(A), middle and figure 2(C), center), but not to the homogenous tumoral construct, that had more structural stability and thus resisted the shock at lower cell numbers, which permitted it to rebound and ‘survive’ (figure 2(A) lower and figure 2(C), right). Kinetic analysis of DCM revealed larger random fluctuations in cell density as well, indicative of a more profound instability of the system than in the previous cases (figure 2(D)).

3.3. Fusion of spheroids in hyperglycemic environments

Raising the glucose flux five times ($5\times$, figure S3) or ten times ($10\times$, figure 3) did not change the pattern of normal spheroids fusion compared to normoglycemic conditions (figures S3(A) and 3(A)), upper and figures S3(C) and 3(C)), left). However, hyperglycemia made even clearer some of the previously observed features when tumor spheroids were involved: (a) the contact of a normal spheroid with a tumoral one led to the eventual engulfment of the

former (figures S3(A) and 3(A)), middle); (b) contacting tumor nodules naturally went first through a fusion/expansion phase, but this invariably reversed, and the pair ended up with a lobular distribution, suggestive of the structural instability seen before (figures S3(A) and 3(A)), bottom).

Higher glucose values in the medium led to increased cell proliferation (figures S3(C) and 3(C)), which resulted in larger fused spheroids. But this size increase came at the cost of subsequent faster glucose use at the construct’s core (figures S3(B), 3(B)), by temporarily generating there more cell death, until the consecutive system shrinkage re-equilibrated it at a smaller scale. This oscillatory pattern also occurred in the early phases of spheroid fusion (until approx. 500 MCS; see figures S3(D) and 3(D)) and compare with figure 1(D)). Of note, despite the increased availability of glucose, the fusing pairs containing tumor spheroids displayed overall reduced glucose levels (as compared with the starting of the process, figures S3(B) and 3(B)), middle and bottom).

DCM analysis also captured the dynamics of spheroid combinations (figures S3(D) and 3(D)), by providing additional details regarding the comparison of individual simulations to each other in the three-fold replicates having more similar courses than for normal glucose concentrations.

3.4. Monitoring binary spheroids fusion by measures of pair geometry

An alternative approach previously used to empirically track the process of spheroids fusion is to measure a fusing pair’s length and width [9]. We applied this method to the homogenous doublets of

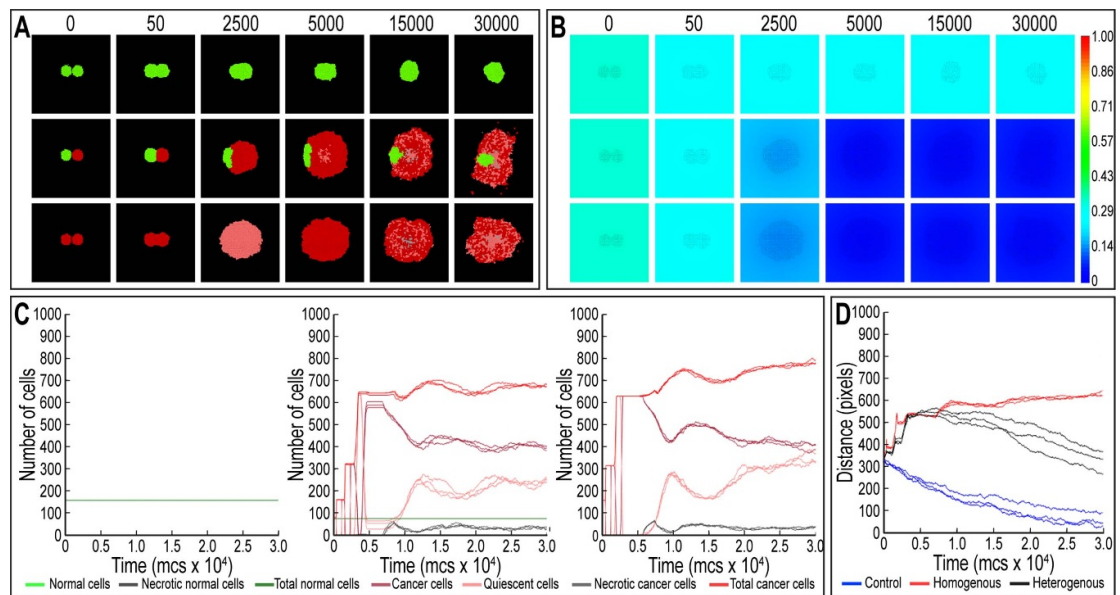


Figure 3. Fusion of cell spheroids at 10 $\times$ glucose concentration. (A) Illustration of fusion process between two normal (upper row), heterogenous (central row) and two tumoral spheroids (lower row). (B) Corresponding glucose fields. (C) Time course of cell subpopulations in the fusing complex (left: normal spheroids; middle: heterogenous pair; right: homogenous tumor spheroids) (color code as in A). (D) Monitoring fusion via the distance between spheroids centers of mass.

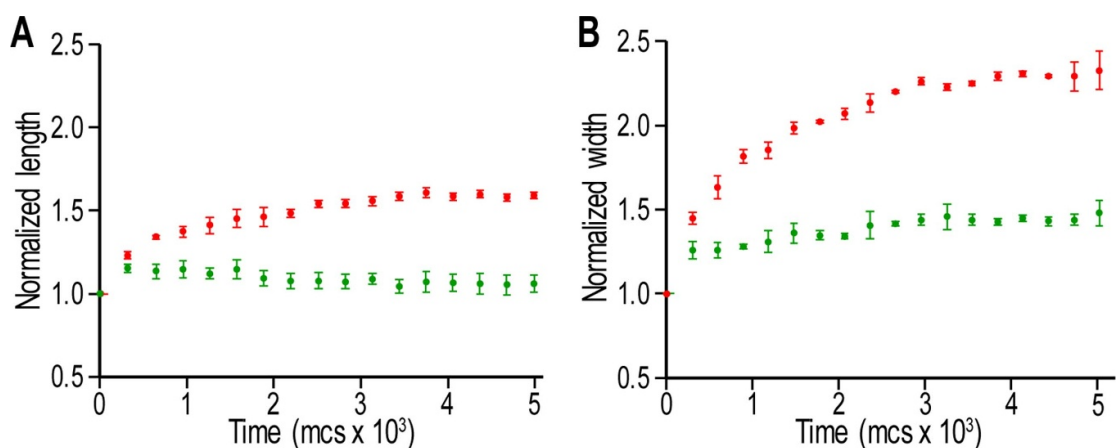


Figure 4. Monitoring binary spheroids fusion at normal glucose levels by measures of pair geometry. (A) Time course of doublets length over an equivalent interval of 24 h (see the text for estimation). (B) Time course of simultaneous doublets widths change over the same interval (color code in both graphs: green, normal cell spheroids; red, tumor cell spheroids). The data represent means $\pm$ SD ($n = 3$). In both cases the normal and tumor curves are statistically different ($p < 0.05$), except for the first two time points. Comparisons at the other glucose concentrations are provided in supplementary data.

normal or tumor spheroids (since the mixed combination is more difficult to interpret from this standpoint). Our simulations applied to normal glucose concentrations showed that the length of normal spheroid doublets started to decrease after contact in a quasi-linear fashion, attaining approx. 15% reduction (figure 4(A), lower curve) in the considered interval. Given the conserved number cells in this normal situation, the width of the construct increased correspondingly with time (figure 4(B), lower curve). In the tumoral case, both the lengths and widths of the doublets increased with time with a parabolic pattern (figures 4(A) and (B), upper curves).

A more extended exploration of fusion kinetics that included longer time intervals and all glucose concentrations used in this study, is presented in figure S4. The doublet length analysis captured the main features of these dependences, such as the stronger impact of glucose availability on tumor than on normal cells. This analysis also confirmed that at very low glucose concentrations the tumor doublets undergo a catastrophic collapse, and at high glucose display the previously noted oscillatory growth in the early phases of fusion (figure S4A).

The doublet widths across all glucose values were also consistent with the kinetics of doublet lengths, and with the imaging and DCM data, e.g. in capturing

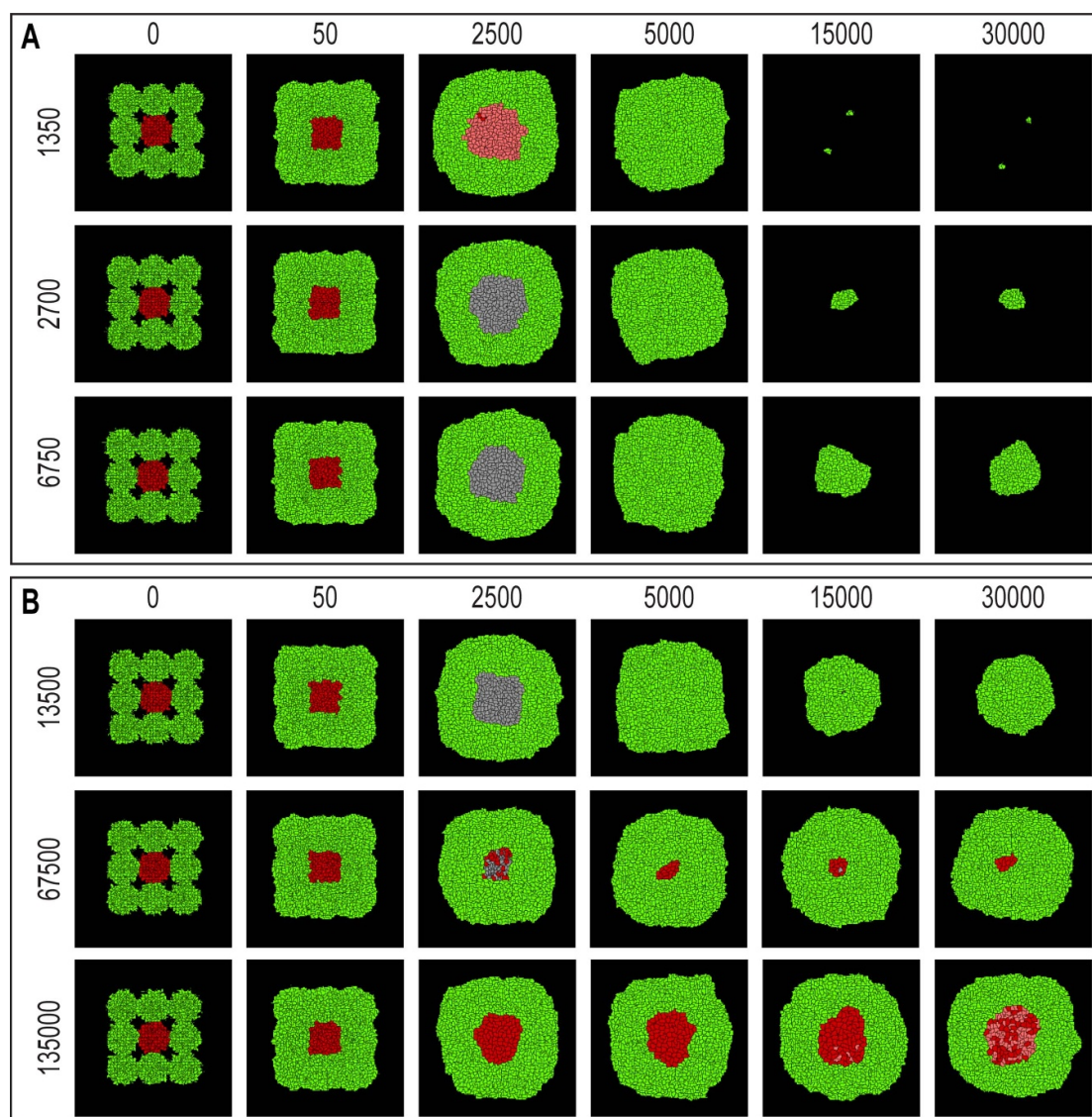


Figure 5. The model of a biofabricated tumor microenvironment from fusing spheroids. (A) Testing the model's evolution at lower than normal ($0.1\times$; $0.2\times$; and $0.5\times$) glucose in the media. (B) Behavior of the complex tumor model at normal ($1\times$) and higher ($5\times$ and $10\times$) glucose in the media. 'Cadherin' (cell-cell interaction strength) values were set to 15 for normal cells, and to 8 for tumor cells. Vertical labels indicate glucose values.

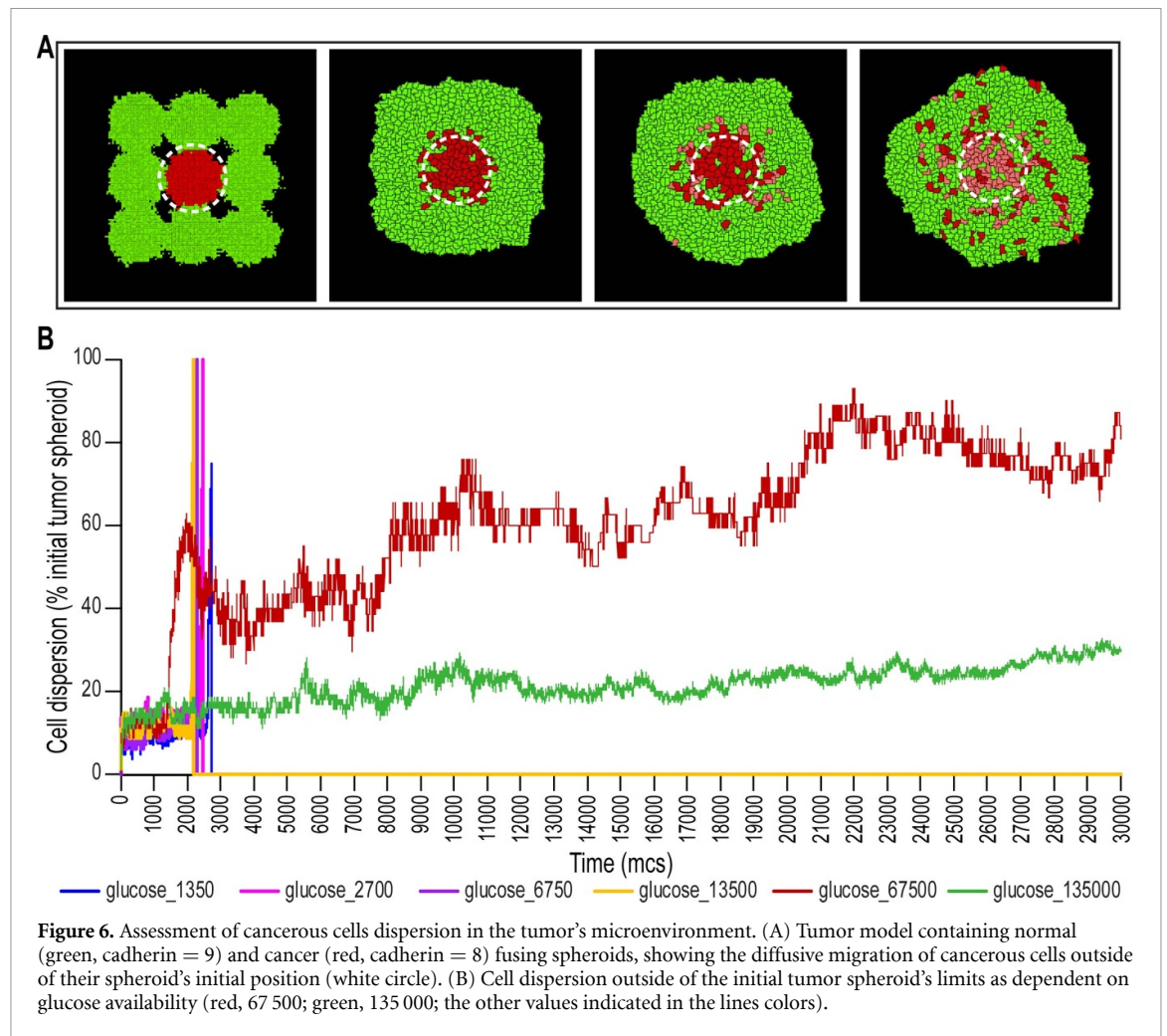
the oscillatory pattern at high glucose concentrations, and identifying the timing of more abrupt transitions. The width changes were even more pronounced in change, as the relative increase in this dimension being about twice as large than the length (figure S4(B)).

Additional insights in the fusion process could be derived from the kinetics of spheroids lengths and widths ratio, a surrogate of doublet's ellipticity (figure S5). The difference between normal cell spheroids and tumor nodules in this regard was highest at normal and quasi-normal glucose availability, with homogenous tumor spheroids elongating back after the initial fusion, due to their tendency to disintegrate. At higher glucose values the tumor nodules were more stable, but early on in the process these tumor doublets tended to round faster than the control ones, while at low glucose concentrations the curves of

the doublet aspect ratio were essentially overlapping (figure S5).

3.5. Post-fusion organization of multiple spheroids in a tissue engineering tumor model

As an application of the CC3D modeling to the scaffold-free bioprinting of a tumor model containing the 'microenvironment' [4], we further simulated the glucose-dependent evolution of a set of eight normal spheroids initially surrounding a tumor nodule (figure 5). Our results show that the assembling of the external spheroids (aligned similarly with their placement for the bioprinting method called 'Kenzan' [10]), generates a continuous cellular 'box' simulating the normal tissue (figure 5). We found that providing the system with glucose at basal or lower than physiological rates, constantly leads to the tumor disappearance (figure 5(A)) even at 'normal' glucose



generation rate (figure 5(B), left). However, in 'hyper-glycemic' conditions the tumor has enough nutrients to become stabilized (figure 5(B) center, right), as also quantified by monitoring the constructs' cellular composition (figure S6).

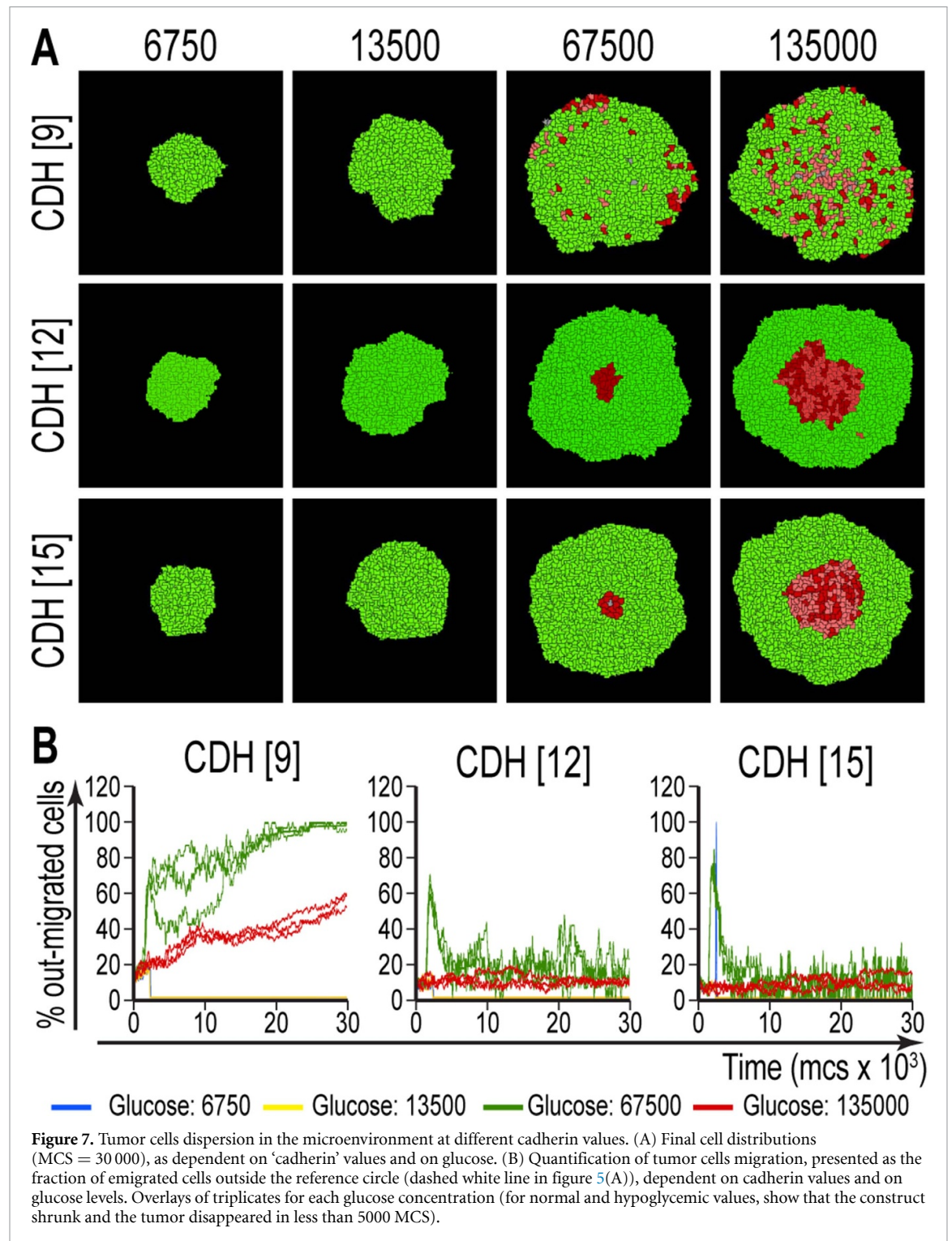
The diffusive migration of the tumor cells within the construct was monitored by determining those able to exit from a circle with the original spheroids' diameter and dynamically positioned on the tumor's center of mass (figure 6(A)). 'Dispersed' cells were considered those located outside a circle whose area is the sum of all tumor cell areas and is centered at the center of mass of all tumor cells at that instantaneous step. To confirm the visual observations, we also analyzed the cellular composition of the constructs (figure 6(B)). We found that this dispersion was dependent both on the 'tightness' of the aggregation of normal cells making the microenvironment, and on glucose availability. Interestingly, at higher-than-normal glucose concentrations the tumor cells could escape from their initial spheroid and spread within the nearby normal tissue even when the intercellular adhesivity between normal cells (cadherin = 9) was slightly stronger than that between the tumor cells (cadherin = 8) (figure 7(A)). Also unexpectedly, the percentage of cells found in the microenvironment

at the highest glucose flux was smaller than at the lower one, because it provided more nutrient to the center of the construct, a place where otherwise the survival of the tumor cells was more metabolically limited. As shown by cell counting, this effect also occurred at higher cadherin values as well, but was of shorter duration, since the cells emigrated from the original spheroid disappearing due to the metabolic competition with the normal cells (figure 7(B)).

4. Discussion

Tumor cells spheroids are valuable *in vitro* tumor models [27], and thus naturally their interaction with other cell spheroids has been studied experimentally on several occasions, mostly as 'confrontation assay' [28]. However, to our knowledge the *in silico* modeling of tumor spheroids fusion has not been reported yet. It is useful to anticipate the behavior and the properties of such non-vascularized constructs, where the impact of limited metabolites availability is even more consequential for cell survival and organization than in binary fusing normal cells spheroids.

Moreover, scaffold-free biofabrication, and in special bioprinting, became a powerful tissue engineering tool, generating more functionally advanced



constructs than the bioprinting based on hydrogels [29]. To take into consideration the contribution of the microenvironment to tumor biology, other cell phenotypes (fibroblasts, endothelial cells, inflammatory cells, etc.) could be added to these 3D cellular constructs. In this case, computational ('virtual') modeling becomes a convenient complement to experimentation [30].

Besides the kinetic aspect of the spheroids fusion, the incorporation of potential regulatory factors such as extracellular glucose has only begun to be

addressed, as we have previously shown for oxygen [21]. In the current study, we extended the approach to include tumor spheroids, and glucose as their metabolic regulator. The opportunity was provided by the open-source CC3D model of a generic tumor spheroid's cellular dynamics [14]. In this model, the availability of glucose was implemented as its 'production rate'. This provides a better analogy with the actual *in vivo* situation where the glucose is continuously refreshed from circulation, than the continuously decreasing concentrations due

to consumption encountered during *in vitro* experiments, as also explained in [14].

4.1. Binary spheroid fusions

The main findings during our simulations of normal and/or tumor spheroid fusion were: (a) when the ‘normal’ cell spheroids and tumor nodules are placed in contact they collapse in larger, generally round structures, as described previously using the same CC3D modeling platform [25]; (b) however, the fused tumor spheroids tend to remain more ellipsoidal, and also to occasionally go in the reverse towards a binary disintegration, except at the highest glucose concentrations (which could be relevant for the facilitation of cancer progression by diabetes); (c) due to the shape fluctuations and structural instability, the monitoring of fusion through shape-related properties for longer intervals was impractical; for this reason, we introduced the DCM, a valuable alternative to track the fusion. This new tracking tool confirmed the observation that, depending on glucose availability (and as the combined result of cell proliferation, death and intra-tumor emergent stratification and evolution [14]), fusing tumor nodules may not irreversibly lead to their aggregation in one single larger spheroid; (d) for longer running times, and particularly at lower-than-normal glucose availability, a construct instability is manifested by stronger random fluctuations of DCM. Such a ‘noise’ analysis of spheroid behavior could be empirically confirmed by fusing spheroids *in vitro*, and used as an independent predictive variable of the effects of various pharmacological treatments.

Furthermore, to compare our results with published experimental observations on fusion of normal and of tumor spheroids [9], we also generated time-course graphs of doublet lengths and widths as dependent on glucose availability, as well as for their ratio, an approach that is more suitable to analyze the empirical data in the early phase of the process (before the structural instabilities described here start to manifest).

The evolution of normal doublets lengths was similar to, and in the equivalent interval of, the *in vitro* experiments presented in [9] (which was 24 h, corresponding to about 5000 MCS if we assume an interval of 20 s per MCS, as in [14]). In the tumoral case, both the lengths and widths of the doublets increased with time, but without the initial reduction in length found for tumor spheroids in [9].

Simultaneously, in our simulations the width of the fusing pair increased, unlike the experimental case in [9], where a shrinkage of the normal spheroids was noted, possibly due to an active constriction involving the cell cytoskeleton as in [31], where ‘chondrospheres’ were used similarly to the chondrocyte spheroids studied in [9]. Another source of discrepancy between the experiments and simulations

could be the three-dimensional (and possibly asymmetrical) nature of the former situation.

Of note, the tumoral fusions for all glucose values became stabilized in about 4000 MCS, before entering a second phase of disintegration. Considering the approximations on the time calibration in [14], this is an equivalent of about 15 h, again in good agreement with the empirical data presented in [9].

The fusions of heterogenous normal-tumor spheroids showed the shrinking of the overall constructs as expected, but also of the tumoral component *per se*, an observation with potentially practical implications, as the *in vitro* interaction of two cell spheroids of similar or different cellular composition is becoming a more common research practice. For example, such experiments are used as ‘confrontation assays’ to study tumor invasiveness [28], microvascular primordia development [32], or the mixing between different cell types [33]. In one of these instances, it was noted that the presence in the same culture well of a normal (osteocyte) spheroid together with one made from breast cancer cells (to simulate bone metastasis) led to an unexpected reduction in the size of the tumor spheroid as compared to the situation when only tumor cells were involved [8]. To explain this phenomenon, the investigators detected a compound secreted by the osteocytes which seemed to inhibit these cancer cells proliferation, but our simulations suggest that the tumor shrinking could have a metabolic component as well.

4.2. Complex tumor models

Cell spheroids have long been used as building blocks in scaffold-free tissue engineering [34]. The basic step in making larger tissue constructs is spheroid fusion, which was explored both experimentally [9] and computationally, including on the CC3D platform [25]. To create more complex *in vitro* tumor models than the currently used single tumor spheroids, we would suggest to take advantage of the spheroids’ fusion propensity, incorporating a tumor ‘core’ into a ‘shell’ made from normal tumor associated cells. This would consist of a cubical ‘box’ composed of 26 normal spheroids distributed on three layers, the middle containing at its center a tumor nodule. This model can be easily implemented using the scaffold-free bioprinting method ‘Kenzan’, which consists of skewering cell spheroids in microneedles [10]. Thus, tissue ‘boxes’ can be created to fully contain both the tumor and its microenvironment (fibroblasts, endothelial cells, inflammatory cells, extracellular matrix, etc.) [4]. After placement in contact, these spheroids would follow the compaction described here, thus permitting the study of similar processes expected to occur in their natural counterparts. However, as our simulations were 2D, here we analyzed the time course of only a section through the middle of this cube. While the results were in part anticipated, e.g. the phenotypic stratification of the cancer cells within the

tumor, favoring the evolution of more diffusive cells [14], or the confining effect of the compacting normal spheroids that may restrain the tumor cells diffusion, others were unexpected. For example, at higher glucose values this constraint was overcome even when the normal cells interacted stronger than the cancerous ones, possibly a cooperative effect resulting from the combination of tumor cells migration and proliferation. Moreover, ‘hypoglycemic’ levels of glucose lead to tumor death, once more confirming the relevance of metabolism for tumor progression, even in a simplified model like ours.

4.3. Other heuristic bio-medical implications of the simulations

Our modeling study provides insights into the role of metabolism in tumor progression that can be verified experimentally and further pursued translationally, despite its several limitations (discussed below). One is the finding of a substantial effect of glucose on spheroid dynamics, even when the cadherin-mediated cell–cell interaction that determines the self-assembling process itself did not depend on glucose concentrations. This was consistent with the insensitivity of the fusion progress dependence on the available glucose in the early phases of normal or tumor spheroid fusion (up to about 500 MSC), but subsequently the proliferative activity and cell turnover increased, and thus altered more substantially both the environment and the tumor’s structure and dynamics.

In addition to the standard (or ‘normal’) glucose flux in the system resulting from constant ‘generation’ and variable ‘consumption’ parameters taken from literature in [14], we also selected different concentrations of this nutrient to mimic hypoglycemic states produced by moderate ($0.5\times$) or severe ($0.2\times$) ‘fasting’, as well as ‘starvation’ ($0.1\times$), and hyperglycemia at two levels ($5\times$ or $10\times$ of the normal, roughly corresponding to diabetes of type II and I, respectively).

In general, our modeling reproduces the anticipated effects of hyperglycemia on tumor growth. However, the effect of glucose on DCM separation does not seem to be proportional with its concentration. Of note, the post-fusion tumor destabilization as found in all instances increases the surface-to-volume (in our 2D case, the length-to-area of the disc) ratio, and thus the chance for single cells to leave the structure, migrate, and form metastases [14]. Remarkably, here we also detected a non-linear loss of tumor resistance to hypoglycemia both regarding the glucose concentrations, and the duration of the exposure. These cooperative effects are characteristic to the systems with complex organization, which are best captured by the ABM type of modeling, as performed here.

Regarding the reduced glucose availability, the nutritional deprivation (‘fasting’) is increasingly used

for dietary/metabolic manipulation of cellular systems. The emphasis has been put so far on its effects on the intestinal microbiome [35], but also on the development of tumors [36], with observations of a possible effect of limiting energy intake on cancer evolution [37]. Our results confirmed the impact of limited glucose availability on tumors development and stability, including the possibility to ‘starve them to death’. While direct experimentation in this area is more difficult for practical and ethical reasons, even with animals, *in silico* simulations emerge as an alternative approach to obtain the needed preliminary information. The selected combinations of parameters can then be more efficiently tested *in vitro* in 3D cultures and in animals.

This modeling could also be used to simulate the effects of ‘intermittent fasting’, assumed to be even more potent than long intervals of food deprivation [38], and thus to optimize the dietary or pharmacological intervention for maximal effect on tumor shrinkage. This line of investigation is particularly promising given the large-amplitude oscillations we found here in cell number and composition, spontaneously occurring in the simulated tumor spheroids, which could be intentionally counter-synchronized with glucose supply for possible tumor eradication.

Our tumor model incorporating a microenvironment demonstrates its strong dependence on glucose availability as well. In low glucose, the tumor might be literally wiped out due to nutrient deprivation. At high glucose concentrations however, although the construct may still shrink, the remnant tumor cells demonstrated an increased intermixing, which could push them in a metastasis-like behavior. In our scenario, such a wound-like ‘healing’ of a tumor by the tumor-associated fibroblasts may be considered an attempt to contain it [39], but this ‘boxing’ could fail in hyperglycemic glucose concentrations, even if the tumor is non-perfused, due to the tumor cells selected *in situ* to survive and then to escape the confinement (as anticipated in [14]).

4.4. Limitations

Among the limitations of this model, as of those from all the GGH class, is the ultra-simplified cellular structure and behavior, yet sufficient for capturing essential features of cellular dynamics. Another is the 2D nature of the simulation, although this still has a heuristic potential, as others [14, 15, 21] and we [21] have shown previously. In addition, the current model has not been spatially or temporally recalibrated, because this has already been originally done in [14], and similarly in [25]. Of note, time calibration differs depending on the method applied, the MCS being shorter when based on cell translocation (as in [14], the model used here as well), than on measurements on 2D images of actual fusing spheroids [25]. The cell–cell interaction strengths and

their variations on a large domain (e.g. doubling ‘cadherin’ values range for normal cells in the microenvironment model), remain arbitrary, but might be considered meaningful for the current modeling purposes.

Regarding spheroid initialization, due to the inherent randomness in the spheroid growth process, the initial number of cells in spheroids had slightly different absolute values (between 76 and 81), although the initial diameter and their positioning of their center of spheroids was the same.

Our model of normal spheroids being a modification of the one devised for tumor case only, we ‘muted’ the tumoral properties starting from the first cell division, in order to maintain the cell volume comparable with that of tumor cells. However, this left in place an initial step of cell volume doubling in normal spheroids also, which is visible when comparing the first two images of the fusing normal spheroids. In addition, this was reflected in an apparent ‘jump’ in the doublet lengths at the beginning of their tracking curves (due the fact that the normalization needed to be done to the doublet length value at time zero). This doublet length–width analysis is an option with its own limitations, since the more complex morphological changes of tumor spheroids occurring later in the fusion process and leading to non-spherical shapes or structural disintegration, cannot be discerned only by these two orthogonal dimensions.

Finally, while the DCM, proposed in the present manuscript, is a valuable metric of model tissue spheroid fusion, it is not straightforward to use it for experimental data analysis. For example, in wide-field fluorescence microscopy images, one could identify pixels associated to each spheroid individually, but in the case of brightfield images of tumor spheroid doublets (which are often irregular and more diffuse) it is not so clear how to compute the DCM. Moreover, DCM analysis alone cannot distinguish compaction with/without cell mixing from their death or proliferation, all being proportionally reflected in changes of DCM values. However, the comparison of these DCM graphs with the visual representations of fusing spheroids helped us to better understand their meaning, and thus to clarify some of the more complex situations.

5. Conclusion

Altogether, our work highlights novel heuristic possibilities of ABM with CC3D of cell spheroids, with significant bio-medical implications which could be further tested experimentally. In particular, even in these very simplified models of a tumor, the relationship of its development with diet and diabetes is transparent, in some respects being possible to compare it with the clinical experience.

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